

Chapter 3

Major plant constituents and their botanical sources

Primary and secondary Metabolites

primary metabolites: it is directly involved in normal Growth, development and reproduction.

Example: carbohydrate, protein, fat and oil, alcohol e.t.c.

- Can be found in almost all organisms and does not possess pharmacological effects.

Secondary metabolites: are not directly involved in growth, development and reproduction of an organism, but they have an ecological function.

- Can be found in the leaves, stem, root or the bark of the plant depending on the type of secondary metabolite that is been produced.
 - **Example:** Glycoside, tannins, Alkaloids.

PRIMARY METABOLITES VERSUS SECONDARY METABOLITES

Primary metabolites perform physiological functions in the body like intrinsic functions

Directly involved in the growth, development, and reproduction

Formed during the growth phase due to the energy metabolism

Produced in large quantities and extraction is easy

Same in all plants

A part of the basic molecular structure of an organism

Ethanol, lactic acid, nucleotides and vitamins are examples

Secondary metabolites are derivatives of primary metabolites

Play a role in ecological functions

Formed near the stationary phase of growth

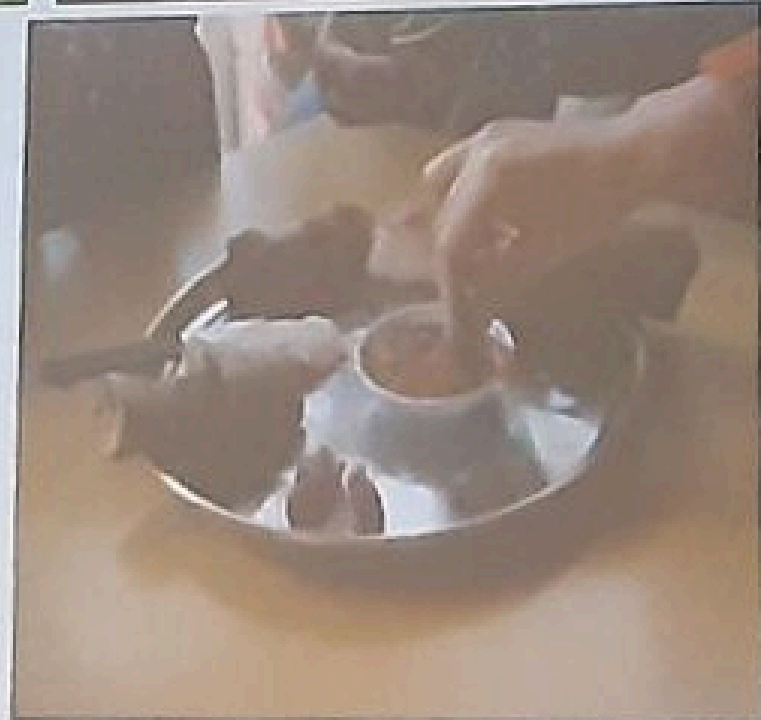
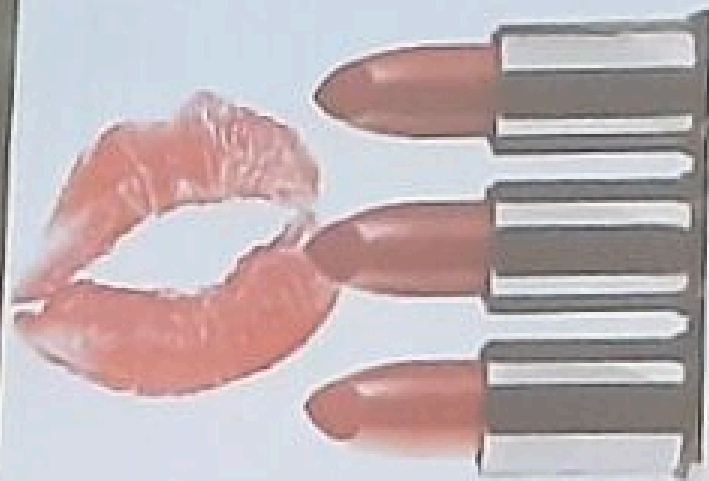
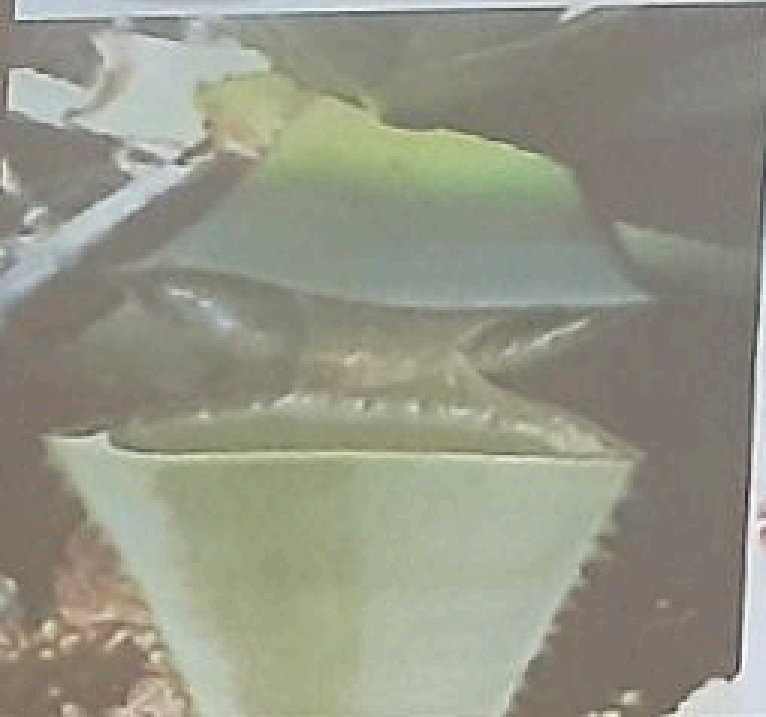
Produced in small quantities and extraction is difficult

Unique to different plant species

Not a part of the basic molecular structure of an organism

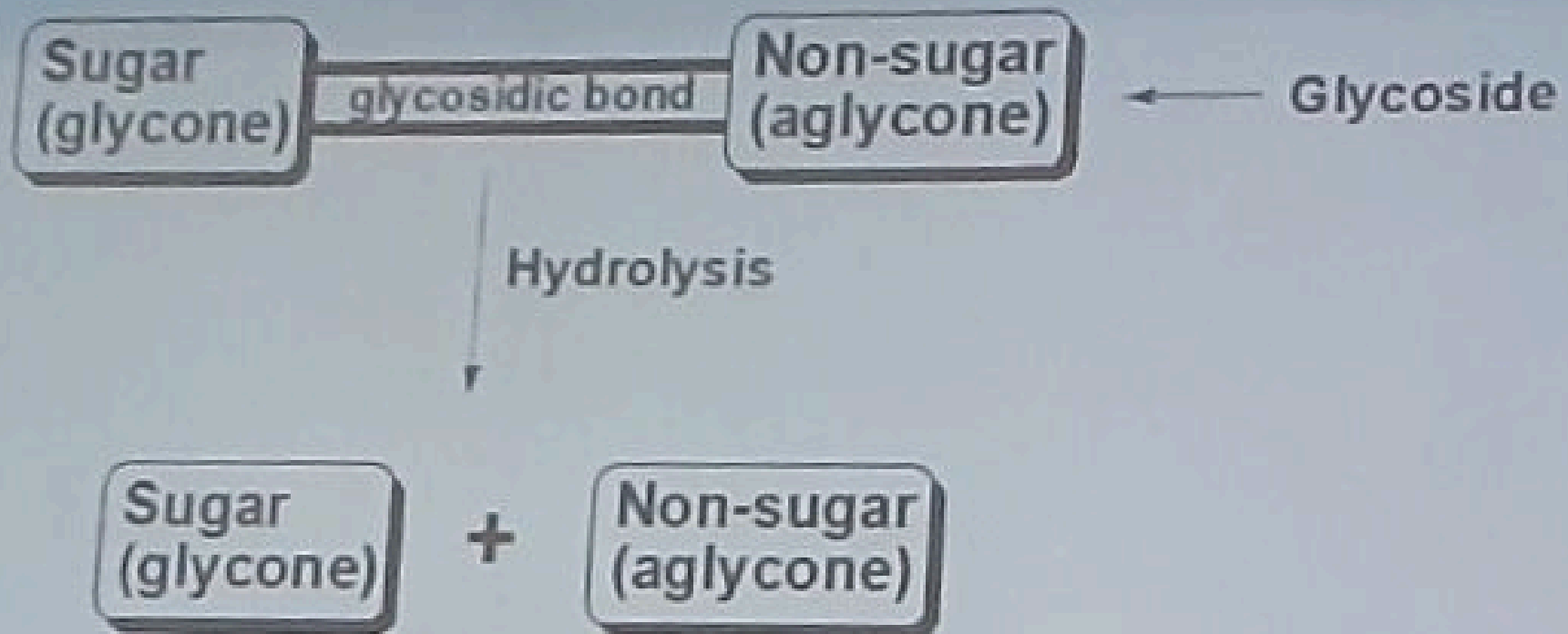
Pigments, antibiotics, and drugs are examples

Glycosides

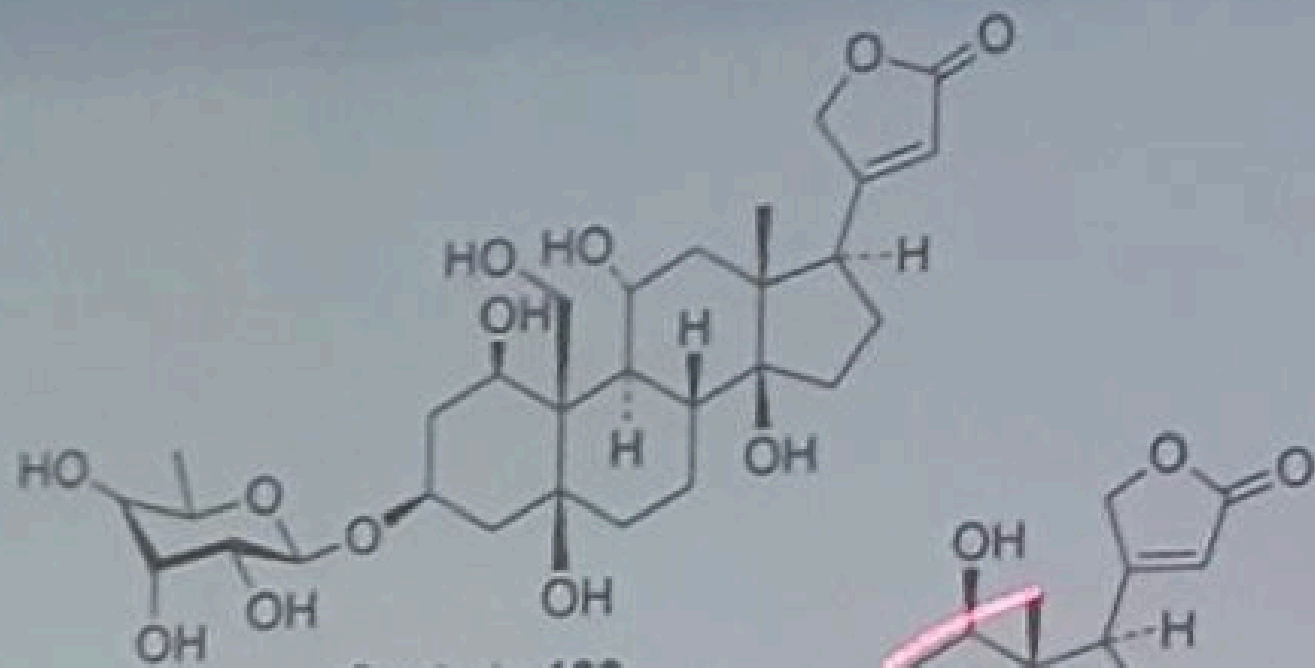


Definition

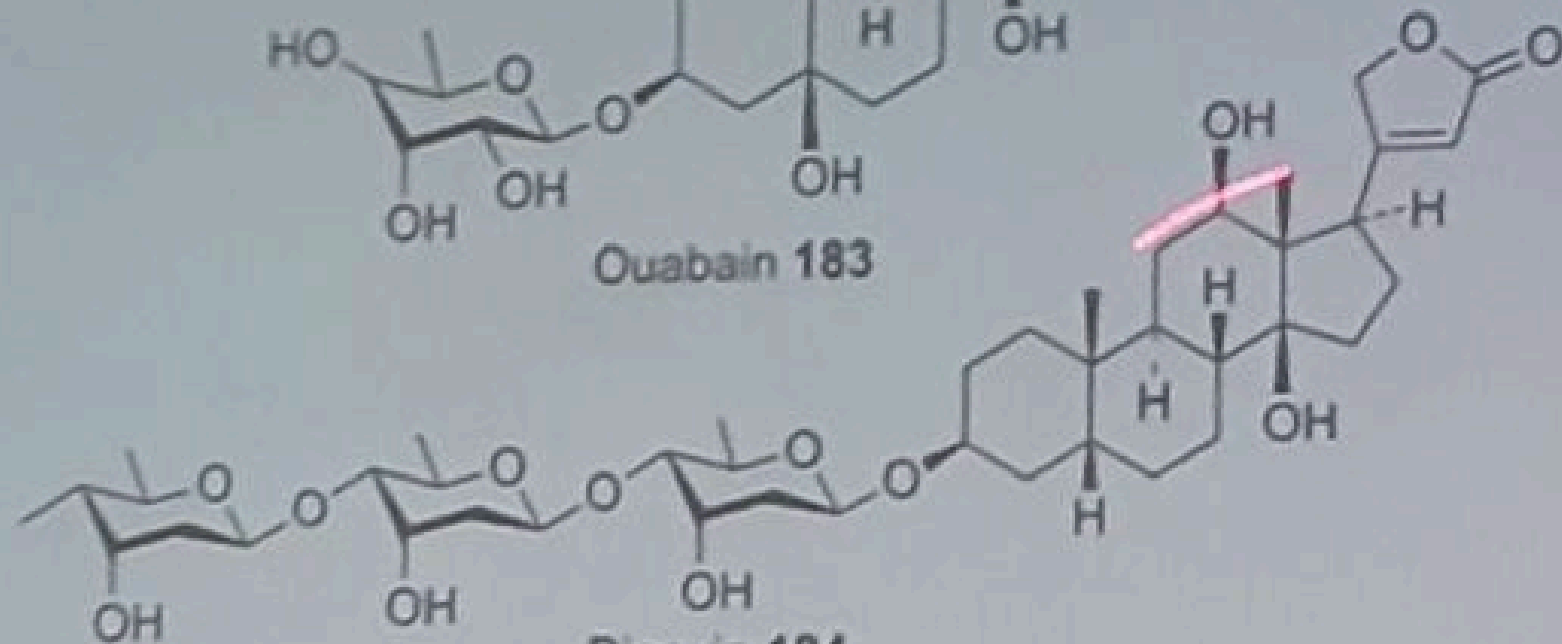
- Is a non-reducing organic compound, usually of plant origin, that is composed of **two chemically and functionally independent moieties**.
- The sugar portion is called **glycone**, while the non-sugar portion is called **aglycone** or **genin**
- Different classes of compounds exist in glycosylated form
 - Eg. Flavonoids, anthraquinones, coumarins...
- Glycosides have long been used as **laxatives, cardiostimulant, analgesic, local irritant, poisons**.



- It's a post modification of the secondary metabolites catalyzed by plant enzymes like **glycosyltransferases**
- **Pharmacological activity** is due to the aglycone
- The sugar part influences **pharmacokinetic**



Ouabain 183



Digoxin 184

Distribution and Function

Distribution

- Plants, Microorganisms, Animals

Function in plants or organism of origin

- Sanitary or protective activities
- Increase solubility of compounds
- Stabilize some unstable compounds
- Regulation of growth and development
- Serve as sugar reserves
- Serve as pollinator attractants
- Important role in signaling
- An allelopathy (biological phenomenon where one plant inhibits the growth of another)

Extraction and Isolation

- Because of the wide range of physical and chemical properties of glycosides and other constituents associated with them
 - no common general method for their isolation is recommended
- Water, methanol, water-ethanol and ethanol are the most common solvents for extraction of glycosides
- Low temperature and neutral conditions are required to avoid decomposition or isomerization.

Precautions Before Extraction

Deactivation of enzymes:

- Drying for 15-30 min at 100 °C followed by slow drying at a low temperature
- Dipping the fresh material into boiling water or boiling alcohol for 10-20 min
- Boiling the fresh plant material with acetone
- Carrying out the extraction at very low temp
- Freeze-drying of the plant material before extraction (lyophilization)
- Carrying the extraction in the presence of $(\text{NH}_4)_2\text{SO}_4$

Precautions Before Extraction

Maintenance of neutral conditions:

- Neutral pH should be assured before and during extraction because: Acidity may result in hydrolysis.
- This is overcome by addition of CaCO_3
- Mild alkalinity may sometimes produce racemization
- Defatting of fat-rich organs (e.g. seeds) before extraction:
 - High amounts of lipoids hinder glycoside extraction
 - usually carried with petroleum ether

Chemical Properties

- Occur as a combination(**glycosilated**) or in free (**aglycone**) form
- Can be a **mono, di, tri saccharide**
 - **D-glucose** is the most common sugar
- They are non-reducing
 - The anomeric carbon is involved in the glycosylation
- Most natural glycosides have **the β -form (more stable)**
- **O-glycoside** is the most abundant type
- Most glycosides are easily hydrolysable

Physical Properties

- Most are crystalline (except saponins)
- Form levorotatory solutions in water and alcohol
- Most have bitter taste
- Solubility

	Water	Alco.	Hyd. Alco	Organic solvents
Glycosides	✓	✓	✓	×
Aglycones	×	✓	✓	✓

Classification:

- Atom from the aglycone involved in the glycosidic linkage:

- | | |
|----------------------|------------------------------|
| • Aglycone- O- Sugar | O-glycosides |
| • Aglycone- C- Sugar | C-glycosides |
| • Aglycone- S- Sugar | S-glycosides (thioglycoside) |
| • Aglycone- N- Sugar | N-glycosides (glycosylamine) |

- Number of sugars:

- | | | |
|----------------|-----------|---------------|
| • One sugar | Monosides | e.g. Salicin. |
| • Two sugar | Biosides | e.g. Diosmin. |
| • Three sugars | Triosides | e.g. Digoxin. |

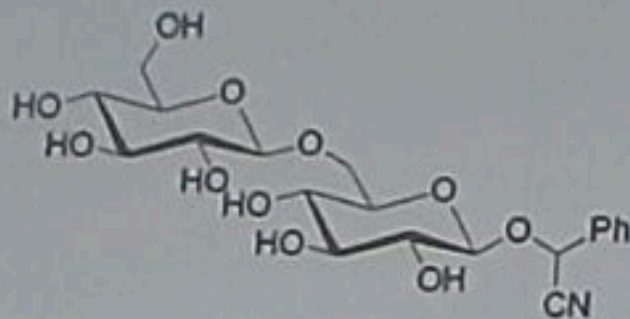
- Nature of the glycoside:

- | | |
|--|-----------------|
| • Primary glycosides: Originally present in the plant | e.g. Purpurea A |
| • Secondary glycosides: Resulted from removal of one sugar from the primary glycosides | e.g. Digitoxin |

- **Type of the glycosidic linkage (stereochemistry):**
 - α - glycosides
 - β - glycosides
- **Source:**
 - Digitalis glycosides
 - Senna glycosides.
- **Therapeutic use:**
 - Analgesic glycosides.
 - Purgative glycosides.
 - Cardiac glycosides.
- **Chemical nature of the aglycone:**
 - Flavone glycosides.
 - Steroidal glycosides.
 - Aldehydic glycosides.
- **Chemical nature of the sugar:**
 - Glucose eg: glucosides
 - Rhamnose eg: rhamnoside

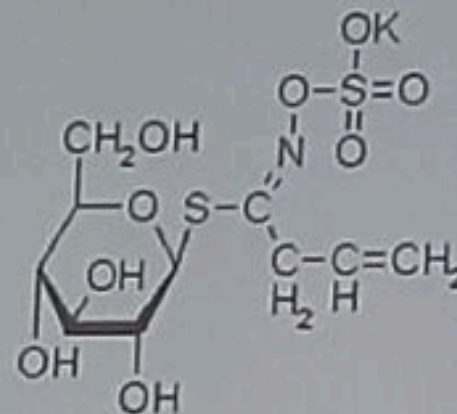
Glycosidic Linkage

O-glycosides



- Are the **most abundant** glycosides
- **Easily be hydrolyzed** by boiling in mineral acids

S-glycosides

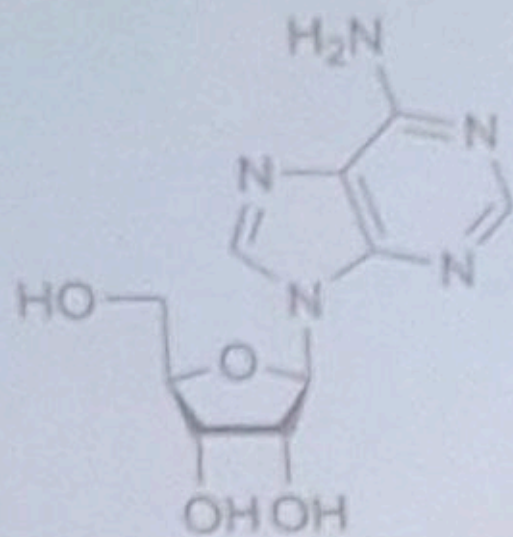


- Natural S-glycosides **are very rare**
- **Easily be hydrolyzed** by boiling in mineral acids

Glycosidic Linkage

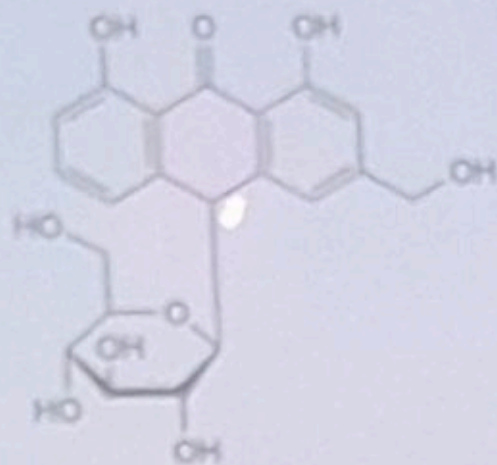
N-glycosides

- easily be hydrolyzed by boiling in mineral acids



C-glycosides

- The sugar is condensed directly to a carbon atom
- highly resistant to acid hydrolysis

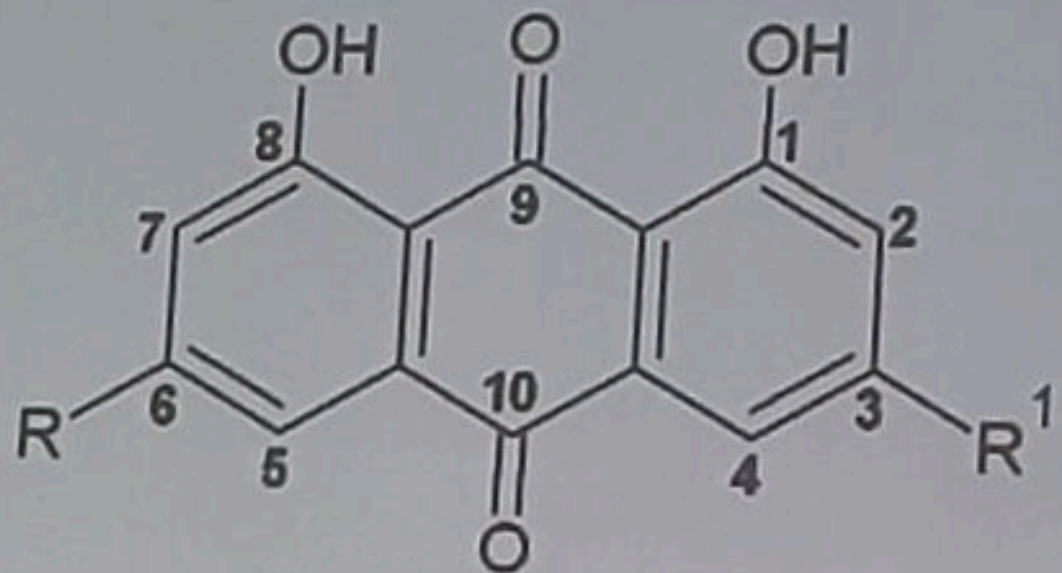


Medicinally and Pharmaceutically Important Glycosides



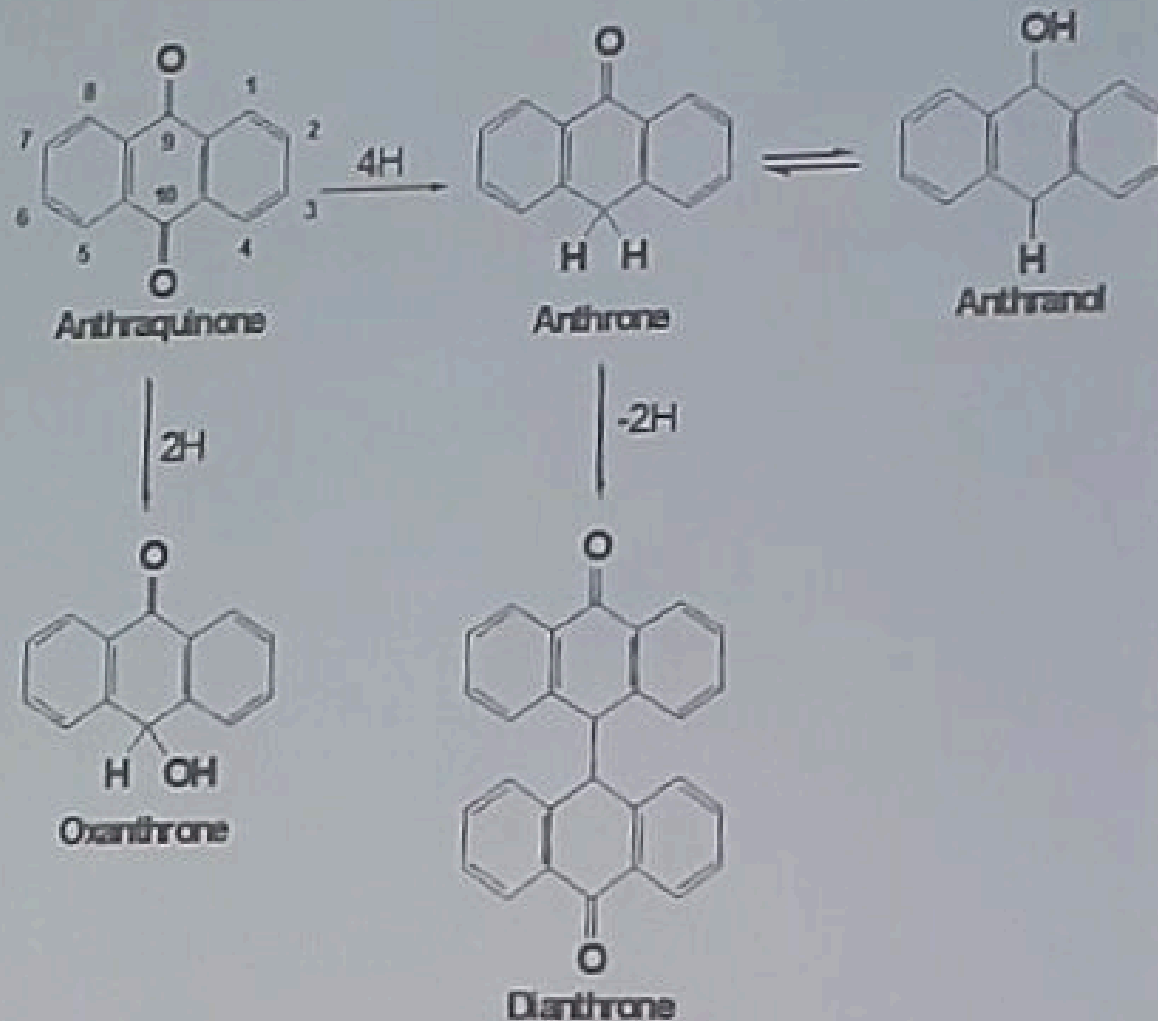
1. ANTHRAQUINONE GLYCOSIDES

- Anthraquinones are naturally occurring structures, which comprise a **central 1,4-diketo-cyclohexa-2,5-diene** (**quinone**) fused to a phenyl ring on either side
- The anthraquinone structure is based on anthracene skeleton

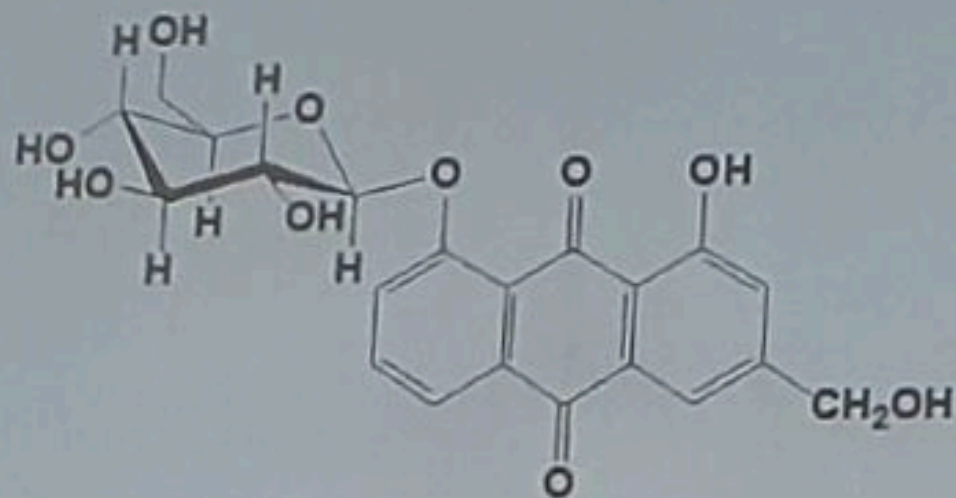


Different derivatives of anthraquinones

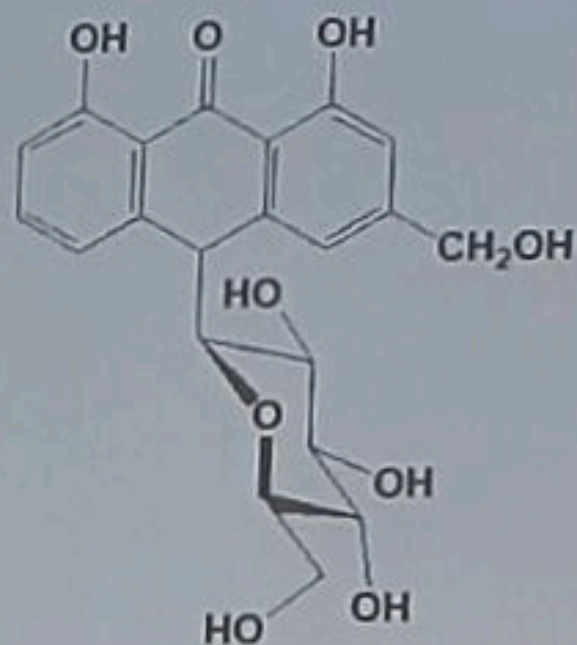
- **Anthrones** (10-H-anthracen-9-ones) and **anthranol**, are reduced forms of anthracene derivatives
- In most anthraquinones **hydroxyl** groups are normally found at positions **C-1** and **C-8**.
- Only **C-3** and **C-6** carbon may also be substituted
- **Not all** anthraquinones are strictly quinones & **Dimerization** is also possible
- Glycosylation at **C-8** or **C-6** hydroxyl groups or **C-glycosides at the C-10**



Examples



Aloe-emodin 8-glucoside
(an O-glycoside)



Barbaloin
(a C-glycoside)

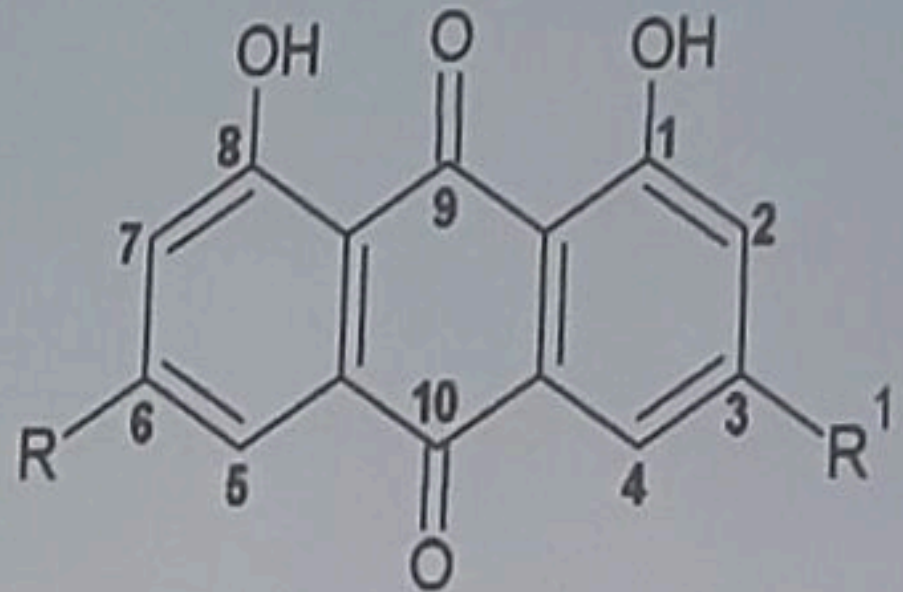
- **Historically:** Rhubarb, Senna, Aloes and Cascara were all used as purgative drugs
- **Monocotyledons:** Only Liliaceae
 - Most commonly C-glycoside: barbaloin
- **Dicotyledons:** Rubiaceae, Leguminosae, Polygonaceae, Rhamnaceae, Ericaceae, Euphorbiaceae, Lythraceae, Saxifragaceae, Scrophulariaceae and Verbenaceae
 - Also in certain fungi and lichen
- Used as laxatives or cathartics, anti-inflammatory, antibacterial, antifungal and also as natural dyes

Pharmacological Activity

- Stimulant cathartics
 - Anthranols and anthrones elicit a more drastic action
 - **The glycosides are hydrolyzed by the enzymes** of the microflora
 - They act on the **large bowel**
 - Their effect may be **delayed up to 6 hrs** or longer
 - **They increase the strength of peristaltic** contractions but not the frequency
 - At larger doses, they cause **colic pain**
 - They **are habit forming**; taken for more than 15 days and cause **excessive loss of fluid and electrolytes particularly potassium.**

Structure-Activity Relationship:

- Glycosylation is essential for activity
- Hydroxylation at C-1 and C-8 is essential for activity
- Oxidation level at C-9 and C-10 is important:
 - Highest level of oxidation have the lowest activity
 - Oxanthrones are less active than anthrones.
 - Complete reduction of C-9 and C-10 eliminates the activity.
- Substitution at C-3 have great impact on activity:
 $\text{CH}_2\text{OH} > \text{CH}_3 > \text{COOH}$



Chemical Tests

Borntrager's test

- Requires free anthraquinone-type aglycones
- **Aglycones produce pink, red or violet color** with ammonium hydroxide solution or sodium bicarbonate solution
- **Glycosides and reduced anthraquinone aglycones** (anthrones, anthranols, oxanthrones and dianthrones) **do not give positive** result in this test
- **Glycosides should be hydrolysed** first by boiling the sample in 2N acid
- **C-glycosides** are hydrolysed by boiling the sample in 2N acid in the presence of **ferric chloride**
- **Oxanthrones** are treated with fuming nitric acid to convert them to anthraquinones

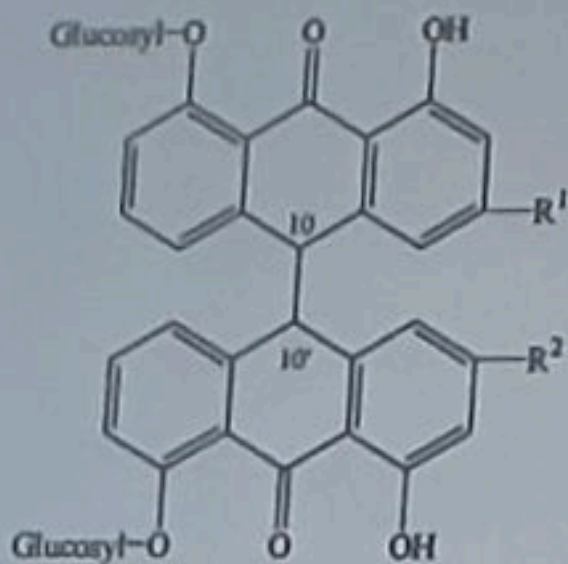
- Schonteten's test (Florence test for anthranols)

- Anthranols produce a **strong green florescence** in borax or other **alkali solutions**.

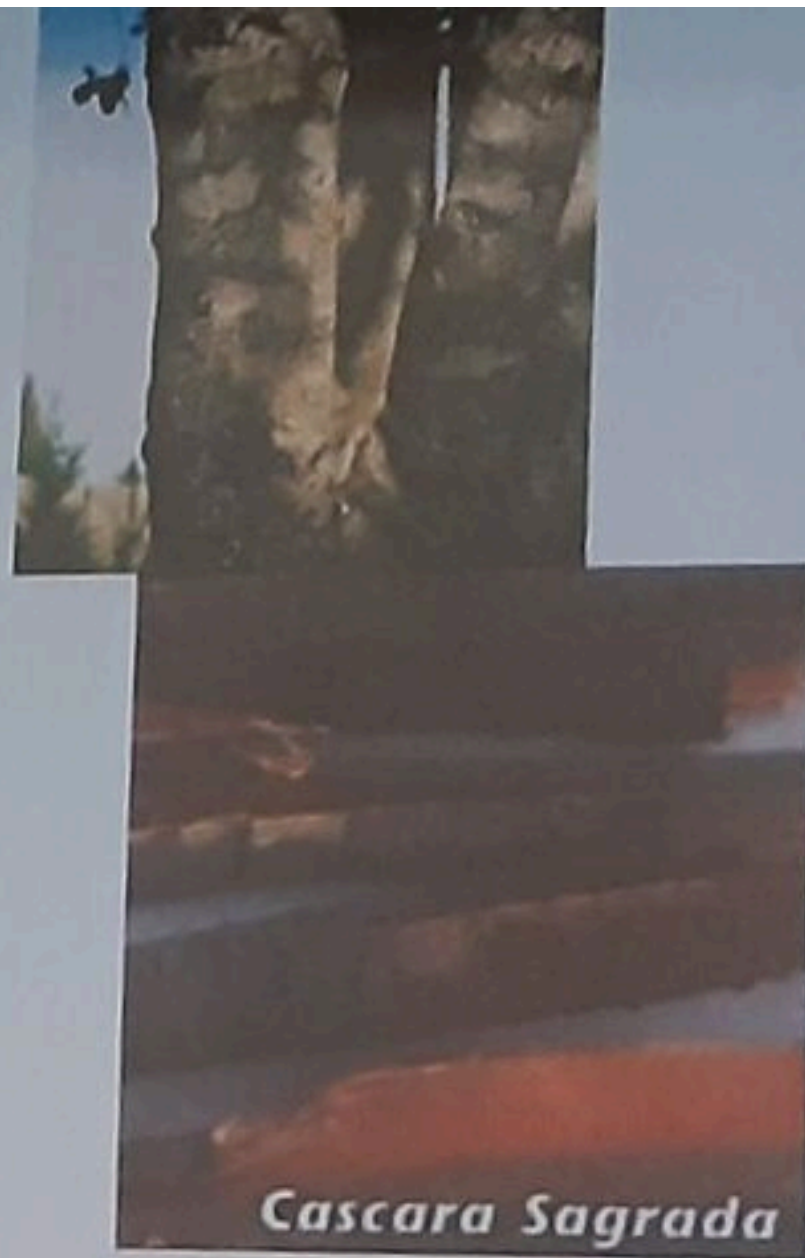
Examples of Crude Drugs Containing Anthraquinone Glycosides

Crude drug	Botanical source	Active constituents	Pharm act.: Med. use
Senna	Dried leaflets of <i>Cassia acutifolia</i> or <i>C. angustifolia</i> (fam. Fabaceae)	Sennosides A and B	Stimulant cathartics
Rhubarb	Dried rhizomes and roots <i>Rheum officinale</i> or other species of <i>Rheum</i> except <i>R. raponticum</i> (fam. Polygonaceae)	Sennosides, chrysophanol glycosides, aloe-emodin glycosides and emodin glycosides	Laxative
Cascara	Dried bark of <i>Rhamnus purshiana</i> (fam. Rhamnaceae)	O-glycosides (10-20%), C-glycosides (80-90%) and some free anthraquinones: e.g. cascariosides	Laxative
Aloe	Dried juice of the leaves of various spp of <i>Aloe</i> like <i>Aloe barbadensis</i> (fam. Liliaceae)	Two fractions: an anthraquinone fraction called "aloin" and resins. barbaloin is the major constituent of the aloin.	Laxative

Cascara



	R ¹	R ²	10-10'
Sennoside A	COOH	COOH	<i>trans</i>
Sennoside B	COOH	COOH	<i>meso</i>

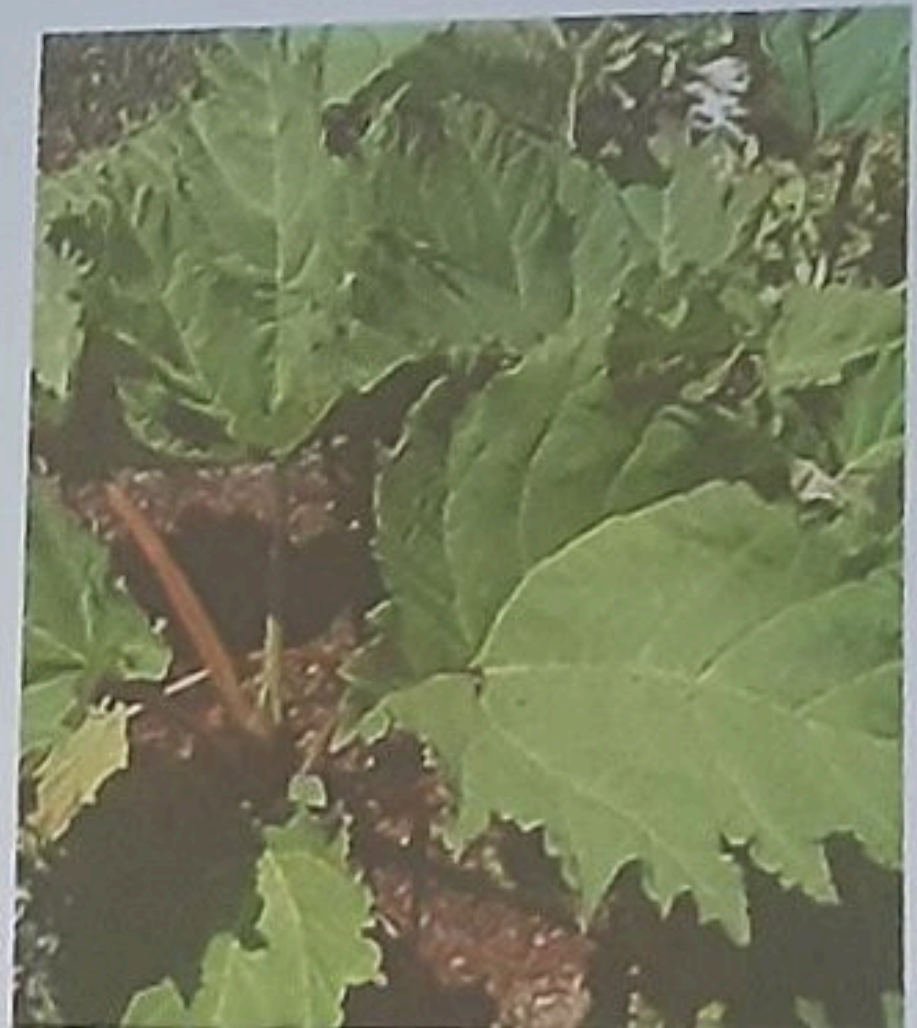
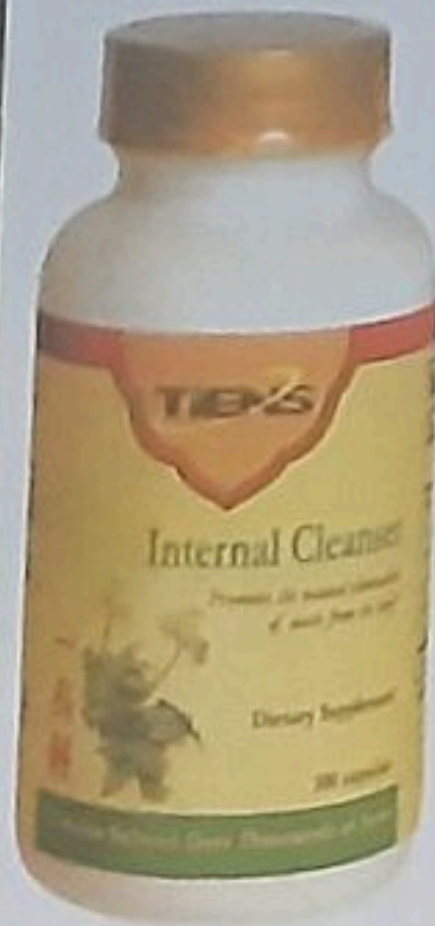
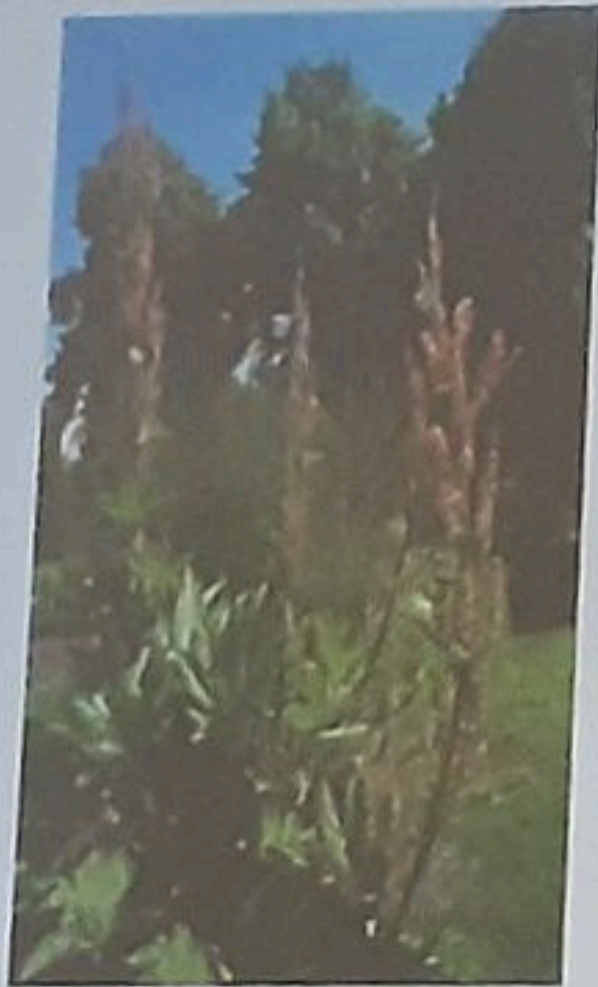


Cascara Sagrada

Senna



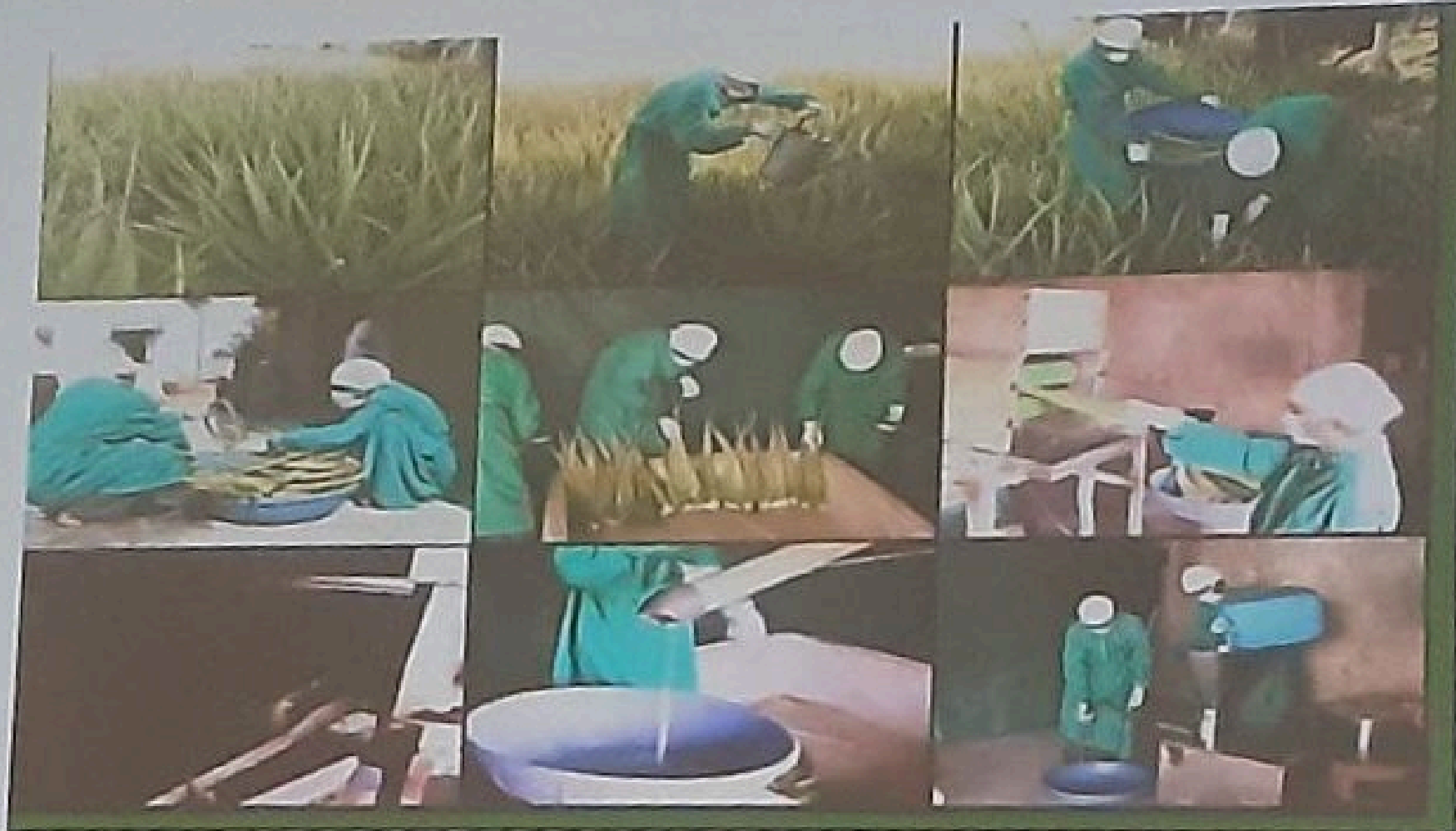
Rhubarb



Aloe



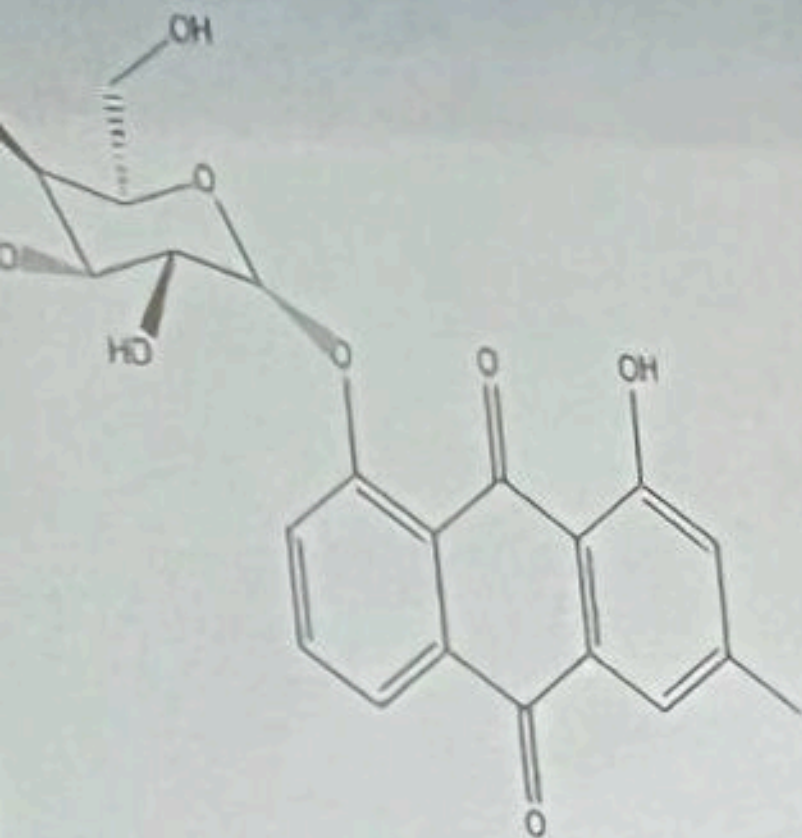
Preparation of Aloes



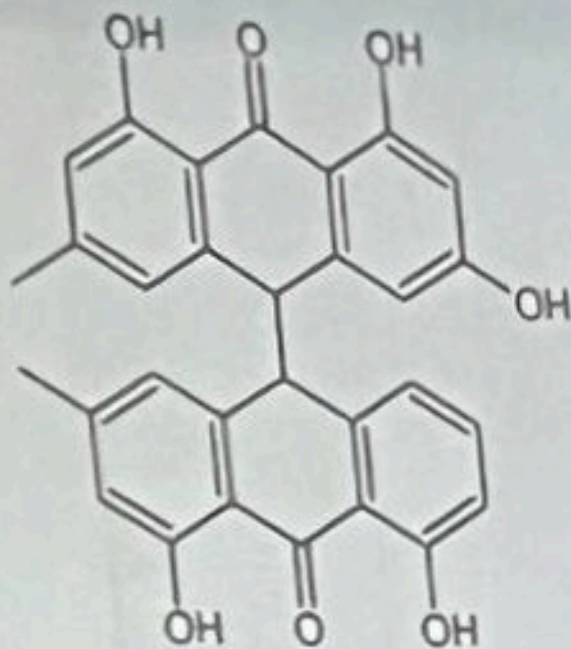
Rumex abyssinicus Jacq (Polygonaceae)

- *Rumex abyssinicus* Jacq (Polygonaceae) has been used traditionally for treatment of hypertension, inflammatory and painful conditions in Ethiopia
- Reported to contain **chrysophanol**, **physcion**, **emodin**, **palmidin C (emodin-chrysophanol bianthrone)**, **chrysophanol-8-O- β -D-glucoside** and **emodin-8-O- β -D-glucoside**

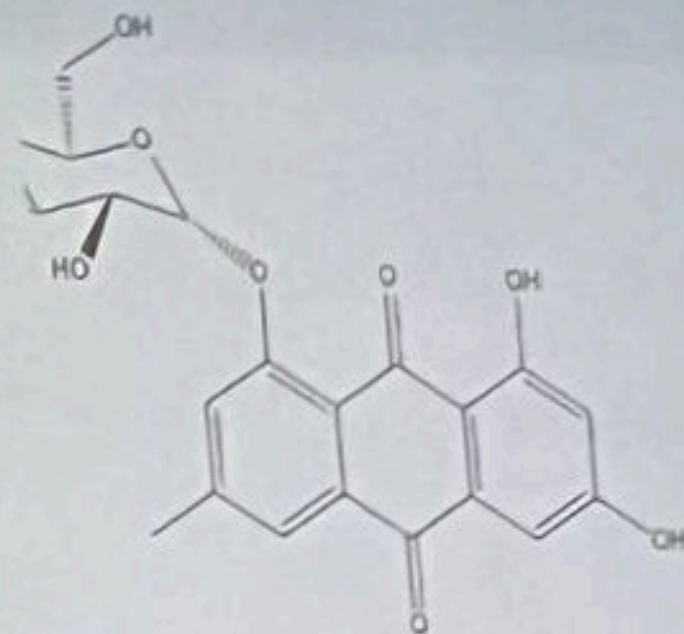




chrysophanol-8-β-D-glucoside



Palmidin C

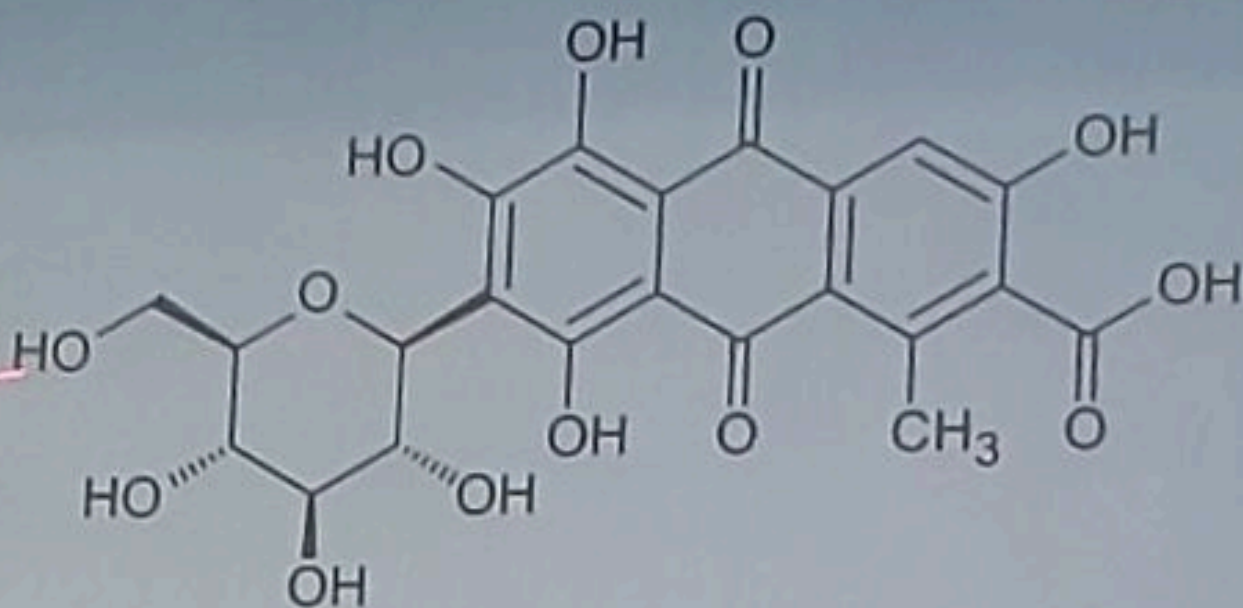


emodin-8-β-D-glucoside

Carminic acid

- Cochineal (*Dactylopius coccus Costa*) infest on cactus pear (*Opuntia ficus-indica*)
- The female insect produces carminic acid to deter predators
- Carminic acid, typically 17-24% of dried insects' weight, can be extracted from the body and eggs, then mixed with aluminium or calcium salts to make carmine dye
- Carminic acid is a red dye used in food, cosmetic, pharmaceutical industries.





9 Foods Made With Crushed Bugs

Starbucks has announced they are reviewing alternative food
 items to boost their (and our) health. But a Strawberry Frost
 isn't the only food you've eaten consumed by crushed bugs.

